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*The Development of Sēgôl in an Open Syllable as a Reflex of *a: An Exercise in Descriptive Phonology*

1. INTRODUCTION

The development of *sēgôl* in an open syllable as a reflex of **a* in the MT¹ is typically treated in Hebrew grammars in two separate categories. In the stressed syllables of segolate nouns it is generally ascribed to the assimilation of **a* to the following (anaptyctic) *sēgôl*. In unstressed syllables (where *sēgôl* usually occurs before a historically lengthened guttural followed by *qāmeṣ*), there is less unanimity on its origin, but the most common view is that it arises through dissimilation of **a* from the following vowel.² This categorization appears to have led to a general opinion that *sēgôl* in an open syllable reflecting **a* arose only under the influence of the following sounds. The vowelizing of the interrogative pronoun *mh* raises difficulties for this view which have not been solved (see BL §33j/p. 266, GB pt. 1 §28a/p. 152, and Joüon §37c/p. 88; the latter notes that the relationship of *mh* to the following word is an important factor). The present article argues that a careful

1. The vowels of the MT are identified in this paper by name. They are transliterated as *u, o, ɔ, a, ε, e, i, ə, ʔ, ā, ē*. The use of *w* as a vowel letter with *u* or *o*, and of *y* with *ε, e*, or *i*, is indicated by a circumflex accent. Other vowel letters are ignored. The (reconstructed) vowel from which an MT vowel arose is represented by the appropriate letter marked with an asterisk. Unstarred vowel letters are used to represent vowels treated in other contexts. Thus **a* in **kasp(u)* gives rise to *ε*, MT *sēgôl*.
2. This *sēgôl* is treated as due to dissimilation in BL §21n-o/p. 216, and most recently by J. Blau, "Marginalia Semitica III," *IOS* 7 (1977) 17 §4 and n. 14. Joüon §29f/p. 74 regards it as the result of "harmonisation vocalique." GB pt. 1 §27c/p. 151 considers the cause uncertain. For their opinions on stressed *sēgôl*, see BL §16d/p. 200, Joüon §96Ab/p. 235, GB pt. 1 §23f/p. 136. For a new suggestion on the development of *sēgôl*, see W. Randall Garr, "The *seghol* and *segholation* in Hebrew," *JNES* 48 (1989) 109–16.

description of the reflexes of **a* in these three situations shows that all can plausibly be explained as resulting from the same process of development, in which the influence of the sounds following **a* is of limited importance.

2. THE REFLEX OF **a* IN SEGOLATE STRUCTURES

2.1 The loss of final vowels (case or mood markers) left a number of words consisting of, or ending in, a structure of form **CaCC*. Where the syllable remains doubly closed, the reflex of **a* is *pataḥ*, the usual reflex of **a* in a closed nonfinal syllable. Typically, however, the final consonant cluster is broken up by the insertion of an anaptyctic vowel: *ḥireq* following *yôd*, *pataḥ* before *hē*³, *ḥêt*, or *ʿayin* and usually after them, *sêgôl* elsewhere.

2.2 The original vowel, which retains its original stress in the now open syllable, appears as *qāmeṣ* where the word stands in terminal position.³ This represents the expected development of **a* in an open stressed syllable.

2.3 Where a word of form **CaCC* stands in medial position, the reflex of **a* is *qāmeṣ* before *wāw*; but *pataḥ* before *yôd*, *hē*³, and *ʿayin* and usually before *ḥêt*; and *sêgôl* elsewhere. The view that this *sêgôl* results from assimilation of **a* to the following anaptyctic vowel fails to explain its appearance where the following vowel is *pataḥ*.

2.4 The general pattern is thus clear. *Pataḥ* appears where the syllable remains closed, *sêgôl* where it is open in medial position, *qāmeṣ* where it is open in terminal position. In medial position, the reflex is further conditioned by the following consonant, so that *pataḥ* and *qāmeṣ* appear as well as *sêgôl*, showing clearly that all three are natural reflexes of **a*. In this context (as in others discussed below) *ḥêt* has a lesser effect on the preceding **a* than does *hē*³ or *ʿayin*.

3. THE REFLEX OF **a* IN THE PRONOUN *mh*

3.1.1 Where *mh* stands at the end of a clause (in terminal position), the reflex of **a* is *qāmeṣ*.⁴

3. "Terminal position" refers to the position in which the intonation contours of the text as read (in the form of the tradition reflected by the vowel patterns) give rise to "pausal" forms, or similar morphophonemic variations. See E. J. Revell, "The Conditioning of Stress Position in Waw Consecutive Perfect Forms," *HAR* 9 (1985) 277-300 §§5-13. The pausal forms of nouns of form **CaCC* typically appear at the ends of clauses. Other positions, where "contextual" forms occur, are called "medial."

4. With *ʔatnāḥ* or *sillūq*, 2 Sam 18:29, Mal 2:14, Ps 84:9, Prov 9:13, Job 13:13; with other disjunctives, Exod 16:7-8, 1 Sam 19:3. Also 2 Sam 18:22-23 (the latter with a conjunctive), where *qāmeṣ* in *mh* could possibly be due to the fact that the following word begins with

3.1.2 Where *mh* stands within a clause, and the following word begins with a consonant other than the gutturals ^ʔ*ālep*, *hē*^ʔ, *hêt*, ^ʕ*ayin*, or *rēš*, the reflex of *a is *pataḥ* where *mh* has *maqṣep*, *səgôl* otherwise.

3.1.3 Where *mh* has *pataḥ*, *dāgēs* is used in the following letter, except where said letter is *yôd* followed by *šewā*^ʔ, or is *l*, *m*, or *š* followed by a consonant with *dāgēs forte*, and in a few anomalous cases.⁵

3.1.4 The pattern of reflexes shown here is the same as that shown in words of form *CaCC and is evidently typical for *a. Minimum change (represented by *pataḥ*) occurs where the syllable is closed (and not originally word final) or is unstressed, medium change (*səgôl*) where it is open in medial position, maximum change (*qāmeš*) where it is open in terminal position.⁶

3.2.1 Where *mh* is followed by a word beginning with *hêt*, the reflex of *a is *səgôl* in eight cases, *pataḥ* in two.

3.2.2 *Pataḥ* occurs in Job 21:21, where *mh* has *maqṣep*, and, where it has a conjunctive, in *ma ḥaṭ:ʕṯî* (Gen 31:36).

3.2.3 *Səgôl* occurs in the parallel to the latter case *ūme-ḥaṭ:ʕṯî* (1 Sam 20:1), where *mh* has *maqṣep*. In the remaining cases, *mh* with *səgôl* has a conjunctive accent. The vowel following the *hêt* is *qāmeš* or *pataḥ*.

3.2.4 The available evidence justifies no more than the statement that the vowel of *mh* before a word beginning with *hêt* is generally similar to the pattern shown before nonguttural consonants.

3.3.1 Where *mh* is followed by a word beginning with ^ʕ*ayin*, the reflex of *a is typically *səgôl*.

3.3.2 *Səgôl* occurs in one case where *mh* has a disjunctive, in twenty-six where it has a conjunctive, and in eleven where it has *maqṣep*. The vowel following ^ʕ*ayin* is *pataḥ* or *ḥaṭṭep pataḥ* in five cases, otherwise *qāmeš*.

⁵*ālep* with *qāmeš*. In an equivocal case in Hag 1:9, *mh* has *səgôl* at the end of a clause, but is followed by the attribution *n^ʔm yhwš šb^ʔwṯ* "(This is) a declaration of the LORD of Hosts." Presumably the prosodic unit ends with this phrase, not with the word *mh*, which is thus not in terminal position. The basis of the division into prosodic units is not simply syntactic but has a wider semantic basis.

5. Anomalous omission of *dāgēs* following *mh* occurs in *y* in Zech 9:17, Job 34:33, Qoh 6:8. On the omission of *dāgēs* in the Leningrad Codex in *l*, *m*, or *š* (and also *q*) following *mh*, see A. Dotan, "Deviation in Gemination in the Tiberian Vocalization," in *Estudios Masoreticos*, (ed. E. Fernández Tejero; Textos y Estudios "Cardenal Cisneros" 33; Madrid: CSIC, 1983) 67. The other Ben Asher codexes rarely agree. *Dāgēs* fails to occur in *yôd* with *šewā*^ʔ in Num 9:8, Judg 7:11, Jer 5:15, Hab 2:1, Ps 85:9.

6. A similar pattern occurs in some nouns of form *CiCC, where the reflexes of *i are *šērē*, *səgôl* (arising from *a* reflecting *i in a stressed closed syllable), and *qāmeš* (the reflex of that *a* in terminal position). See E. J. Revell, "The Tiberian Reflexes of Short *i in Closed Syllables," *JAOS* 109 (1989) 183–203.

3.3.3 The reflex of *a is *qāmeš* in two cases where *mh* has a conjunctive: *mō ʿim:ādī* (Gen 31:32) and *mō ʿabdəkō* (2 Kgs 8:13). The last case has a close parallel in *me ʿabdekō* (2 Sam 9:8), which shows the expected *sĕgōl*.

3.3.4 In this situation, no difference in vowelizing correlates (even generally) with the use of *maqqup* rather than a conjunctive. The two cases in which *qāmeš* occurs could have been partially conditioned by the vowel following the ʿayin, but, if so, other factors must also have been involved.

3.4.1 Where *mh* is followed by a word beginning with *hē*, the reflex of *a is usually *qāmeš*.

3.4.2 The reflex of *a is *pataḥ* before *hūʔ* and *hīʔ* (the only cases where *mh* is followed by an open monosyllable). *Mh* has *maqqup* in six cases, a conjunctive in one.⁷

3.4.3 The reflex of *a is *sĕgōl* before *hōyō* (with final or penultimate stress) and *howē* (the only cases where the word following *mh* is composed of two open syllables). Here also, *mh* has *maqqup* in six cases, a conjunctive in one.⁸

3.4.4 The reflex of *a is typically *qāmeš* where *mh* has *maqqup* but stands before a word in which more than one consonant follows the initial vowel, as *mō-hōʿōm* (Jer 33:24), or where *mh* has a conjunctive accent, as *mō hem* (Ezek 8:6 *Qere*). There are, in all, nearly forty cases.

3.4.5 There is one exception (besides the cases noted above where *mh* with *pataḥ* or *sĕgōl* has a conjunctive). *Mh* with *sĕgōl* has a conjunctive in *me hōʿōdōm* (Qoh 2:12), contrasting with the close parallel in *mō hōʿōšōm* (1 Sam 6:4), which shows the expected vowelizing.

3.4.6 The use of *sĕgōl* and *pataḥ* appears to be determined by the length (in terms of number of consonants) of the word following *mh*, and by the use of *maqqup*. Where the word is too long, or where *maqqup* is not used, *pataḥ* and *sĕgōl* typically do not occur. The choice of *pataḥ* or *sĕgōl* could have been determined by the structure of the word following *mh*, or by its initial vowel (see §4.3.3–4).

3.5.1 Where *mh* is followed by a word beginning with a guttural, the reflex of *a is affected by the vowel following the guttural only to a minor extent, if at all. The main conditioning factors are the use of *maqqup* (as where the following word does not begin with a guttural) and the nature of the following guttural.

3.5.2 The development of *qāmeš* is more likely where the following guttural is articulated more toward the back of the mouth, or is voiced. Thus,

7. *Maqqep* is used in Gen 23:15; Exod 16:15; Num 13:18, 16:11; Zech 5:6; Ps 39:5. A conjunctive occurs in Esth 8:1.

8. *Maqqep* occurs in Exod 32:1, 23; 1 Sam 4:16; 2 Sam 1:4; Lam 5:1; Qoh 2:22. A conjunctive occurs in Qoh 7:10.

where consonants are articulated in the same position, *qāmeš* develops before the voiced ^ˈ*ayin*, but not before the voiceless *hêt*. Where both are voiceless, *qāmeš* develops before the glottal *hēˀ*, but not before the laryngeal *hêt*. *Rêš* is voiced, and ^ˈ*ālep* has glottal articulation, but there is no good basis for a suggestion of the qualities which made these more conducive to the development of *qāmeš* than the three fricatives.

3.5.3 *Pataḥ* occurs before the voiceless fricatives *hēˀ* and *hêt*. Before the glottal *hēˀ* it occurs only under special conditions (to which there is no parallel before *hêt* or before ^ˈ*ayin*). It thus appears likely that these three fricatives are conducive to the development of *səgôl* (which is used elsewhere) in the same degree as to the development of *qāmeš*.

4. THE REFLEX OF *a IN THE DEFINITE ARTICLE

4.1 Before nonguttural consonants, the reflex of *a in the definite article is *pataḥ*. The following letter is marked with *dāgēš* except (in many cases) where it is followed by *šewāˀ*.

4.2.1 Before *hêt*, the reflex of *a is usually *pataḥ*.

4.2.2 Where *hêt* is followed by *qāmeš*, the reflex is *səgôl* where *qāmeš* is stressed or stands in an open syllable, including *hātēp qāmeš*, as *heḥōdōšim*, *heḥōrobôt*, *heḥōrebôt*.⁹ Where *qāmeš* stands in a closed unstressed syllable, the reflex of *a is *pataḥ* in *haḥokmō*.

4.2.3 In a few anomalous cases, the reflex of *a is *qāmeš*, as in Gen 6:19; Isa 3:22, 17:8.

4.3.1 Before *hēˀ*, the reflex of *a is usually *pataḥ*.

4.3.2 Where *hēˀ* is followed by *qāmeš* in an unstressed open syllable, the reflex of *a is *səgôl*.

4.3.3 Where *hēˀ* is followed by stressed *qāmeš* or *šērē*, the reflex of *a is *qāmeš*.

4.3.4 Where *hēˀ* is followed by the pronoun *hūˀ* or *hīˀ*, or by unstressed vowels other than *qāmeš* or *šērē*, the reflex of *a is *pataḥ*.

4.4.1 Before ^ˈ*ayin*, the reflex of *a is usually *qāmeš*. This includes those cases in which ^ˈ*ayin* is followed by stressed *qāmeš* as in *hōˀom*, by *qāmeš* in a closed unstressed syllable (in proper nouns in Lev 11:13, Num 3:27, Deut 14:12, Josh 18:24, 1 Chr 26:23), and by *qāmeš hātūp* in *hōˀōmōrīm* (Ruth 2:7, 15).

9. In the last of these, *hātēp qāmeš* evidently reflects *a. I can find no source showing the form *heḥōlī*, listed by E. König, *Lehrgebäude der hebräischen Sprache* (Leipzig: Hinrichs, 1881), 1:134.

4.4.2 Where ^ˈayin is followed by *qāmeš* in an unstressed open syllable, the reflex of *a is *səgôl*.

4.4.3 In a few anomalous cases, the reflex of *a before ^ˈayin is *pataḥ*. In several of these, this *pataḥ* contrasts with the use of the expected *qāmeš* before other examples of the same word. Thus *qāmeš* occurs before ^ˈenayim in Gen 3:6, 38:21; Prov 10:26; but *pataḥ* in 1 Sam 16:7, Qoh 11:7.

4.5 Before ^ˈālep and *rēš*, the reflex of *a is *qāmeš*.

4.6.1 It is clear that the effect of a following guttural on the *a of the article is much the same as its effect on *a in *mh*. ^ˈālep and *rēš* induce the maximum change, ^ˈayin and *hē* less change, and *hêt* minimum change. Unlike the vowel of *mh*, however, the *a of the article is clearly affected by the vowel following *h*, *ḥ*, or ^ˈ.

4.2.6 Following *hêt*, *qāmeš* is more conducive to change in preceding *a than is any other vowel. *Ḥātēp qāmeš* has the same effect, whether it reflects *a or *u. Only *qāmeš* in a closed unstressed syllable induces the same (minimum) change as do other vowels.

4.6.3 Following *hē*^ˈ, *qāmeš* also produces greater change than do other vowels, with one exception. Where it is stressed, *qāmeš* induces maximum change in preceding *a. The same is true of stressed *sērē*. Where *qāmeš* is unstressed, it generally induces medium change. Other unstressed vowels (including *sērē*), and other stressed vowels, induce minimum change.

4.6.4 Following ^ˈayin, unstressed *qāmeš* in an open syllable (reflecting *a) usually induces only medium change. All other vowels, including *ḥātēp qāmeš*, *qāmeš ḥātūp* (which reflect *u), and stressed *qāmeš*, induce maximum change.

5. THE EFFECT OF A GUTTURAL ON PRECEDING *a

5.1 Gutturals are generally conducive to the change of preceding *a toward *qāmeš*, but this is true of the fricatives *h*, *ḥ*, and ^ˈ only where they are followed by a vowel. Unless the standard pattern calls for *qāmeš* in the final syllable, *h*, *ḥ*, or ^ˈ in word-final position is preceded by *pataḥ* (either regular or “furtive”). The guttural does not induce further development, even where the vowel is stressed. In structures of form *CaCC, where *a is followed by *h*, *ḥ*, or ^ˈ, the vowel develops only minimally in medial position, and so does not appear as *səgôl*, as it does before other consonants. It appears that the development of *a is similarly inhibited by following ^ˈayin where it is followed by *a. Where the following *a is stressed, the expected development could occur, as before any other vowel. Where it is not stressed, however, the change in a preceding *a proceeded more slowly than before other vowels.

5.2 Blau is no doubt right in his view that this effect of **a* following *ʿayin* on preceding **a* occurred while the reflexes of **a* and **u* were phonetically distinct, as this explains the fact that the maximum change in the preceding **a* takes place before *ʿayin* followed by *qāmeš* reflecting **u*, as before any other vowel.¹⁰ There seems no doubt, however, that the changes in **a* followed by a guttural were stimulated by the Masoretic *qāmeš* (ׁ), which might reflect **a* or **u*, not by long *ā*. The failure of the **a* of the article to develop into *sēgōl* in *haḥōkmō* is due to the fact that the following *qāmeš* stands in a closed syllable, not to the fact that it reflects **u*, since **a* does change to *sēgōl* before *ḥēt* followed by *ḥāṭep qāmeš* reflecting **u* which stands in an open syllable.

5.3 The situation of **a* before *ʿayin* followed by **a* is analogous to that of the first vowel in the absolute form *ʾaḥat*. The first **a* is represented by *pataḥ* (rather than by *sēgōl*, as in the masculine form *ʾeḥōd*) because in medial position the effect of the historical double closure of the final syllable in the feminine form prevented the development of its vowel to *qāmeš*. The *pataḥ* in this final syllable, though stressed and (presumably) long in the Masoretic pronunciation, had no effect on the preceding **a*.¹¹ In terminal position, this inhibiting factor is nullified, so that *qāmeš* appears in the second syllable and *sēgōl* in the first, as in the masculine form. In the same way, *ʿayin* followed by **a* has inhibited the development of preceding **a*, unless the vowel following *ʿayin* is stressed. This stress nullified the inhibiting factor (presumably by hastening the development of the stressed vowel toward *qameš*) and so allowed the expected change to take place in the preceding **a*.

6. THE DEVELOPMENT OF **a* IN SEGOLATES, *mh*, AND THE ARTICLE

The survey of the reflexes of **a* in the three common situations suggests that *sēgōl* develops most readily in a stressed open syllable or in an unstressed syllable which was historically closed but is open in the MT, exactly as does *qāmeš*. A guttural followed by a vowel is favorable to the change in preceding **a*: *ʾālep* and *rēš* most, *hēʾ* and *ʿayin* less, *ḥēt* least. The last three are more favorable to the change where the following vowel is stressed *qāmeš*. In this situation, the reflex of **a* is *qāmeš* before *hēʾ* and *ʿayin*, but only *sēgōl* before the less favorable *ḥēt*. Voiceless *hēʾ* or *ḥēt* where followed by

10. Blau, "Marginalia Semitica," 17 n. 14.

11. The probability that the vowel of a stressed syllable is always long is shown by G. Khan, "Vowel Length and Syllable Structure in the Tiberian Tradition of Biblical Hebrew," *JSS* 32 (1987) 44. However, the relation of Khan's sources to the tradition represented in the MT is unclear, so his conclusions must be treated with caution.

unstressed *qāmeš* in an open syllable provides a more favorable environment for change than where followed by some other vowel. This is not the case with the voiced *ʿayin* followed by unstressed *qāmeš* in an open syllable. The change of **a* toward *qāmeš* is retarded in this situation. This explanation is preferable to that of dissimilation, since there is no reason to regard *sēgôl* as a reflex of **a* as resulting from dissimilation in other situations. There is, then, no impediment to seeing the different reflexes of **a* in the situations described as representing different stages in the same process of change. Where conditions for change in such situations are optimal, the reflex of **a* is *qameš*. Where they are less good, the reflex is *sēgôl*.

7. *Sēgôl* IN OTHER MORPHS

7.1 *Prefixed Interrogative Hē*⁷

7.1.1 The reflex of **a* in interrogative *hē*⁷ is *pataḥ* or *ḥātēp pataḥ*, except where the word begins with a guttural followed by *qāmeš* in an unstressed open syllable. In this situation, the reflex of **a* is *sēgôl*. This includes the cases of *ḥēt* followed by *ḥātēp qāmeš* in *heḥḏaltî* in Judg 9:9, etc.¹² I know of no cases where *hē*⁷ interrogative stands before *qāmeš* in a stressed or closed syllable, nor of any in which it stands before *ʿayin* followed by *qāmeš*.

7.1.2 The only exceptions to this pattern appear to be cases where the prefixed *hē*⁷ was taken as representing the definite article, as in Qoh 3:21 and probably Num 16:22, Deut 20:19. The conditioning of **a* in this morph is, then, generally similar to that in the article. The differences are presumably due to the phonological differences between the two morphs.

7.2 *The Prefix š-*

7.2.1 The prefix *š-* is commonly voweled with *sēgôl* before all consonants, including *ʿ, h, ḥ, ʿ*. Except where one of these four follows *š-*, *dāgēš* is marked in the following letter. This includes *rēš* (Song 5:2) and *yôd* followed by *sēwā*⁷ (Ps 123:2).

7.2.2 In a few cases this prefix has other vowels. *Pataḥ* occurs before a nonguttural (which is marked with *dāgēš*) in Gen 6:3, Judg 5:7bis, Song 1:7,

12. This form is most reasonably explained as a perfect *Qal*, with original **a* reduced between long syllables, in GKC §63k/p. 167. On the tendency to reduce **a* before a nonreduced syllable, see J. Blau, *A Grammar of Biblical Hebrew* (Wiesbaden: Harrassowitz, 1976) §9.3.3. The perfect form is used to represent a hypothetical occurrence which is unexpected or undesirable, as *gul:aḥtî* (Judg 16:17) and other cases in 2 Sam 15:33, 2 Kgs 7:4, Ruth 1:12, etc. See S. R. Driver, *A Treatise on the Use of the Tenses in Hebrew* (3d ed.; Oxford: Clarendon, 1892) 178/§ 138i(α).

and perhaps Job 19:29. *Qāmeš* is used before ^ʔ*ālep* in Judg 6:17. *Šēwā*^ʔ is used before *hū*^ʔ and *hem* in Qoh 2:22, 3:18.

7.2.3 Different vowelings may be used in similar situations, as before *gam* (Qoh 1:17, 2:15, 8:14; cf. Gen 6:3) and *hem* (Song 6:5, Lam 4:9; cf. Qoh 3:18). The situations in which the different vowels are used are not absolutely identical (note the accentuation where the prefix occurs before *hem*), but it seems highly probable that the different vowelings are free variants. That with *səgōl* can be considered irrelevant to the particular development of *a traced here. The cases described in § 7.2.2 show a pattern possibly related to that shown by the vowel of the definite article, but minimum development is shown only before *hū*^ʔ, while maximum development occurs before *hem*, as before *hē*^ʔ with stressed *qāmeš*.¹³

7.3 Elsewhere before a Guttural Followed by *Qāmeš*

7.3.1 *Səgōl* occurs before *hēt* (whether the following *qāmeš* is stressed or unstressed) in *Hithpael* forms from *nhl* and *nḥm*, and nouns such as ^ʔ*eh*^ʔ*od*, as listed in GB pt. 1 § 28a/p. 152. In *mibteḥ*^ʔ (Prov 21:22, the final vowel represents a pronoun) and *mibteḥom* (Jer 48:13), where *hēt* was not historically lengthened, the following *qāmeš* has induced greater development than has occurred before other vowels (see Jer 2:37, Ps 71:5, Prov 22:19, Job 8:14).

7.3.2 *Səgōl* occurs before *hē*^ʔ even where the following *qāmeš* is stressed in the *Hithpael* form *hiṭ:eh*^ʔ*rū* (Num 8:7, 2 Chr 30:18), and the nominal form *beh*^ʔ*ret* (Lev 14:56), as well as where it is unstressed in *beh*^ʔ*rot* (Lev 13:38–39).

7.3.3 *Səgōl* even occurs before ^ʔ*ālep* in *ne*^ʔ*ṣōṣōt* (Neh 9:18, 26), although *qāmeš* occurs in *ne*^ʔ*ṣōṣōṭek*^ʔ (Ezek 35:12). *Šēwā*^ʔ appears in *ne*^ʔ*ṣōṣ*^ʔ (2 Kgs 19:3 = Isa 37:3), which could be the singular form of this word but is not usually considered so.

7.3.4 *Səgōl* occurs before *rēš* in a stressed syllable in *he*^ʔ*ro* (Gen 14:10). The expected *qāmeš* occurs in *h*^ʔ*oh*^ʔ*ro* (Gen 12:8, etc.).¹⁴

7.3.5 The reflex of *a before ^ʕ*ayin* is *qāmeš* in forms from the feminine noun *ro*^ʕ (from *r*^{ʕʕ}), whether the following *qāmeš* is stressed or unstressed. *Qāmeš* also occurs in the *Hithpael* form *niš*^ʕ*ro* (Isa 41:23), where the stress is penultimate.

13. S. Morag suggests that the vowelings with *šēwā*^ʔ derives from a different stream of tradition—not from the one in which the use of *səgōl* became standard; see “On the Historical Validity of the Vocalization of the Hebrew Bible,” *JAOS* 94 (1974) 308–9.

14. Where directional *hē*^ʔ is not suffixed to this noun, a more developed reflex of *a is more likely to be shown where the article is prefixed than where it is not, and this is the case also with some other nouns of form *CaC:, as *bad*, *gan*, *ḥag*, ^ʕ*am*. *Pataḥ* occurs in the construct form *midbar*^ʔ (Josh 18:12, 1 Kgs 19:15).

7.4 Other Situations

7.4.1 *Sĕgôl* occurs in an unstressed open final syllable as the vowel of directional *hē*³ in 1 Sam 21:2, 22:9; 1 Kgs 2:36, 42; 2 Kgs 5:25; Ezek 25:13 and as that of a third-person feminine singular perfect verb form in Zech 5:4.

7.4.2 *Sĕgôl* replaces the expected *qāmeš* in a stressed open final syllable in the characteristic ending of a feminine noun in Isa 59:5, in the "cohortative" ending in 1 Sam 28:15, and in the related ending of the imperative in *dā³e ḥākma* (Prov 24:14, the only case where the following word begins with a guttural followed by *qāmeš*).

7.4.3 *Sĕgôl* (presumably reflecting **a*) occurs in a stressed open nonfinal syllable before *nūn* followed by *qāmeš* in the construct form in *pad:enō ʔārōm* (Gen 28:2, 5, 6, 7; cf. §7.3.4), also in the third-person feminine plural suffix in *qirbenō* (Gen 41:21*bis*), and similarly before *kap* and *hē*³ followed by *qāmeš* in the second-person masculine singular and third-person feminine singular pronominal suffixes *-ekō* and *-ehō*.

8. THE DEVELOPMENT OF *Sĕgôl* IN THE OTHER SITUATIONS

8.1 The cases of *sĕgôl* in final open syllables are rare exceptions to the common forms. They cannot be interpreted (in the light of the present evidence) as examples of the process of change described in §6. Of the other cases, *pad:enō* could be explained as an example of this process. This seems less likely for the *sĕgôl* in the pronominal forms. Indeed it is doubtful if it is correct to regard these simply as reflexes of **a* in a stressed open syllable; other factors are probably also involved. The same is true for the use of *sĕgôl* in the prefix *š-*. There is insufficient evidence of the use of other vowels in that prefix to allow a decision.

8.2 The other cases of *sĕgôl* before gutturals (§§7.1, 7.3) are generally consistent with the pattern of development shown by the vowel of the article. They reinforce the conclusion that *pataḥ*, *sĕgôl*, and *qāmeš* represent stages in a single process of development. The development was conditioned by the total phonological environment, not just by the sounds that follow. This would include the structure (in terms of syllable structure, stress pattern, and vowel length) of the prosodic phrase in which the vowel occurs.¹⁵ Consequently, the vowel which actually reflects **a* before a particular guttural followed by stressed or unstressed *qāmeš* differs in different words.

15. On the importance of prosodic phrases (or "speech units") in conditioning vowel and stress patterns, see E. J. Revell, *Nesiga in Tiberian Hebrew* (Textos y Estudios "Cardenal Cisneros" 39; Madrid: CSIC, 1987 §9.1-4; idem, "Conditioning of Stress Position," 299; idem, "Stress Position in Verb Forms with Vocalic Affix," *JSS* 32 (1987) 259.

Sēgōl may occur where the article would show *qāmeš*, and *qāmeš* may occur where the article would show *sēgōl*, showing again that *sēgōl* does not result from dissimilation, but from a less advanced stage of the same process of change which gave rise to *qāmeš*.

9. CONCLUSION

9.1 *Sēgōl* as a reflex of **a* appears most commonly in syllables which have been secondarily opened, but this is not the case with interrogative *hē*¹⁶, nor with some of the noun forms considered, and may not be the case with *mh*. It is probably correct, then, to say that *sēgōl* developed as a reflex of **a* in a syllable in which the vowel remained short at a later date than in the typical open syllable, due either to structural features (the vowel stood in a closed nonfinal syllable) or to prosodic features (the morph was typically proclitic,¹⁶ or the syllable was never stressed). Consequently the process of change (which was due to the opening of the syllable, or to the effect of a following guttural) began relatively late in the history of the language, and so only induced the maximum development (*qāmeš*) in the vowel in situations particularly favorable to the change. *Sēgōl* represents a medial stage in this process of change.

9.2 The anomalies noted in the above descriptions can be accounted for in a number of ways. Studies of various phenomena in the Masoretic Text suggest that the accentuation of that text does not always represent the same intonation patterns as conditioned its vowelings.¹⁷ Thus, when the vowel patterns were fixed, some cases of *mh* now unstressed may have been stressed, and vice versa. The process by which the Masoretes established a standard reading may have resulted in the introduction into the standard text of a form which had developed under conditioning characteristic of some slightly different strand of tradition.¹⁸ Finally, the development of **a* may

16. G. Khan, "The Pronunciation of םה before *dageš* in the Medieval Tiberian Hebrew Reading Tradition," *JSS* 34 (1989) 433–41, shows that *mh* with *pataḥ* and following *dāgēš* was read as proclitic in the Tiberian tradition of the Ben Asher school, but was read with a certain degree of prosodic separation at later times. Possibly this reflects the final stage of a development away from what was once general proclitic use. I am most grateful to Dr. Khan for his kindly providing me with a copy of this article in typescript, and for permission to cite it.
17. See my discussion of this point in "Conditioning of Stress Position," 281–84; and "Stress Position in Verb Forms," 250.
18. This possibility is discussed in E. J. Revell, "LXX and MT: Aspects of Relationship," in *De Septuaginta: Studies in Honour of John William Wevers* (ed. A. Pietersma and C. Cox; Toronto: Benben, 1984) 48. Cf. Morag's opinion on the origin of the use of *šewā*² as the vowel of the prefix *š-* referred to in n. 13 above.

have been affected by prosodic features which are little understood, such as the tendency to alternate long and short syllables noted in n. 12 above.

9.3 The general line of development of **a* in the situations surveyed is, then, quite clear. An open syllable, a following guttural, main stress, and terminal position are all conducive to the development of *qāmeṣ* from **a*. Where all, or most, of these factors are present, **a* gives rise to *qāmeṣ*. Where a few or none are present, **a* gives rise to *pataḥ*. In intermediate situations, **a* gives rise to *səgôl*. The effect of these factors on the development of **a*, and so the number required to effect development, varies with the individual features which characterize the different situations surveyed.

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